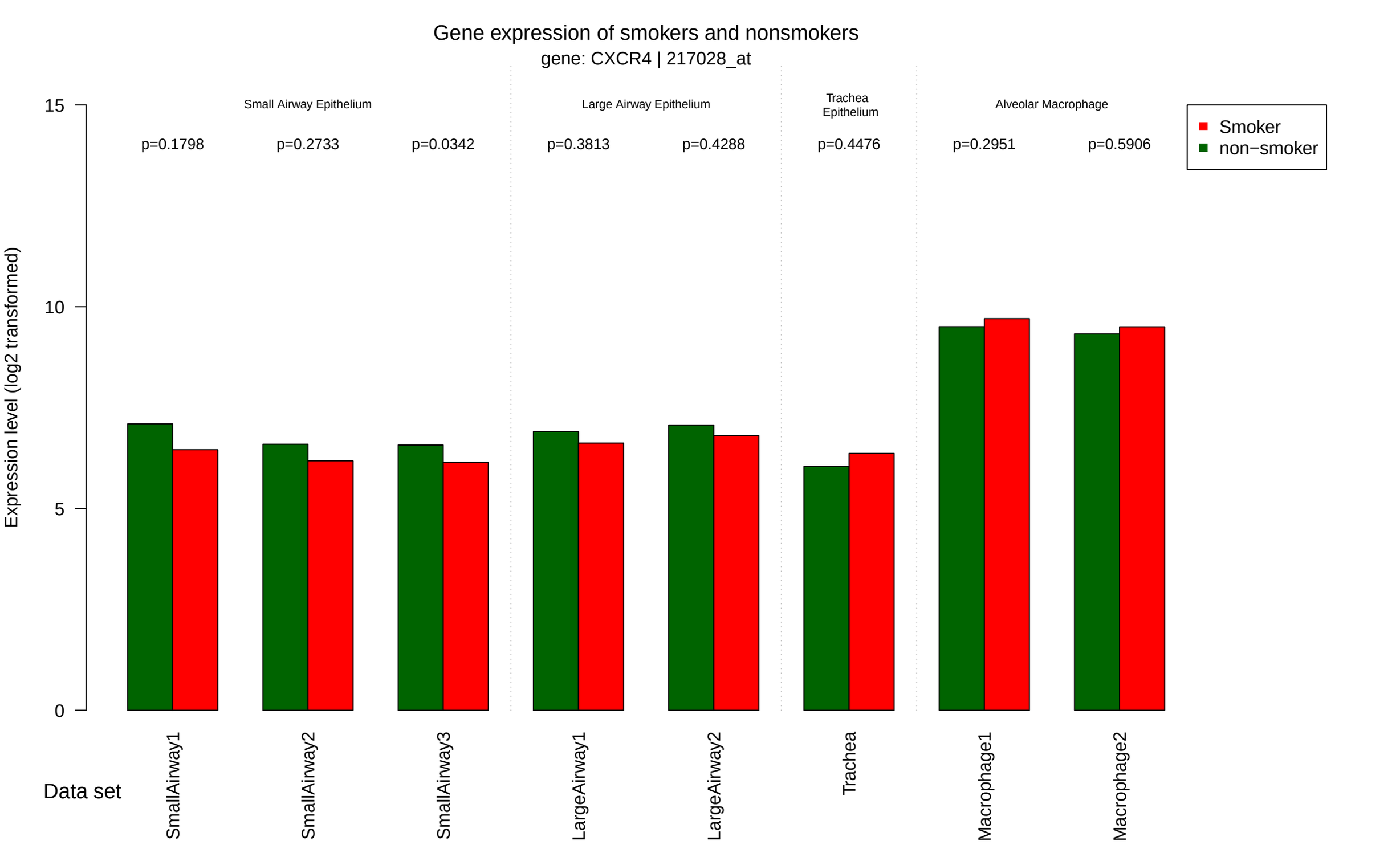
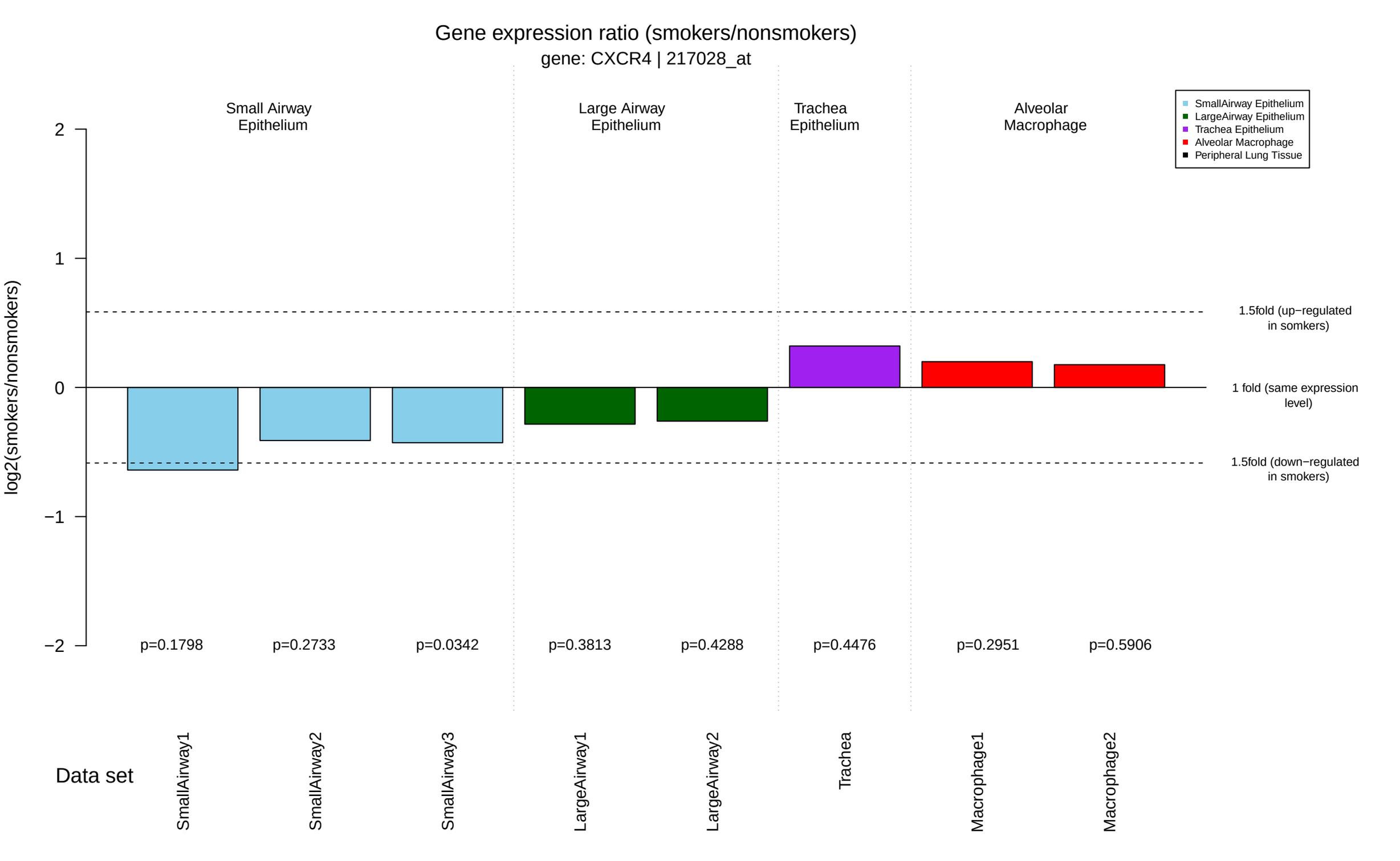
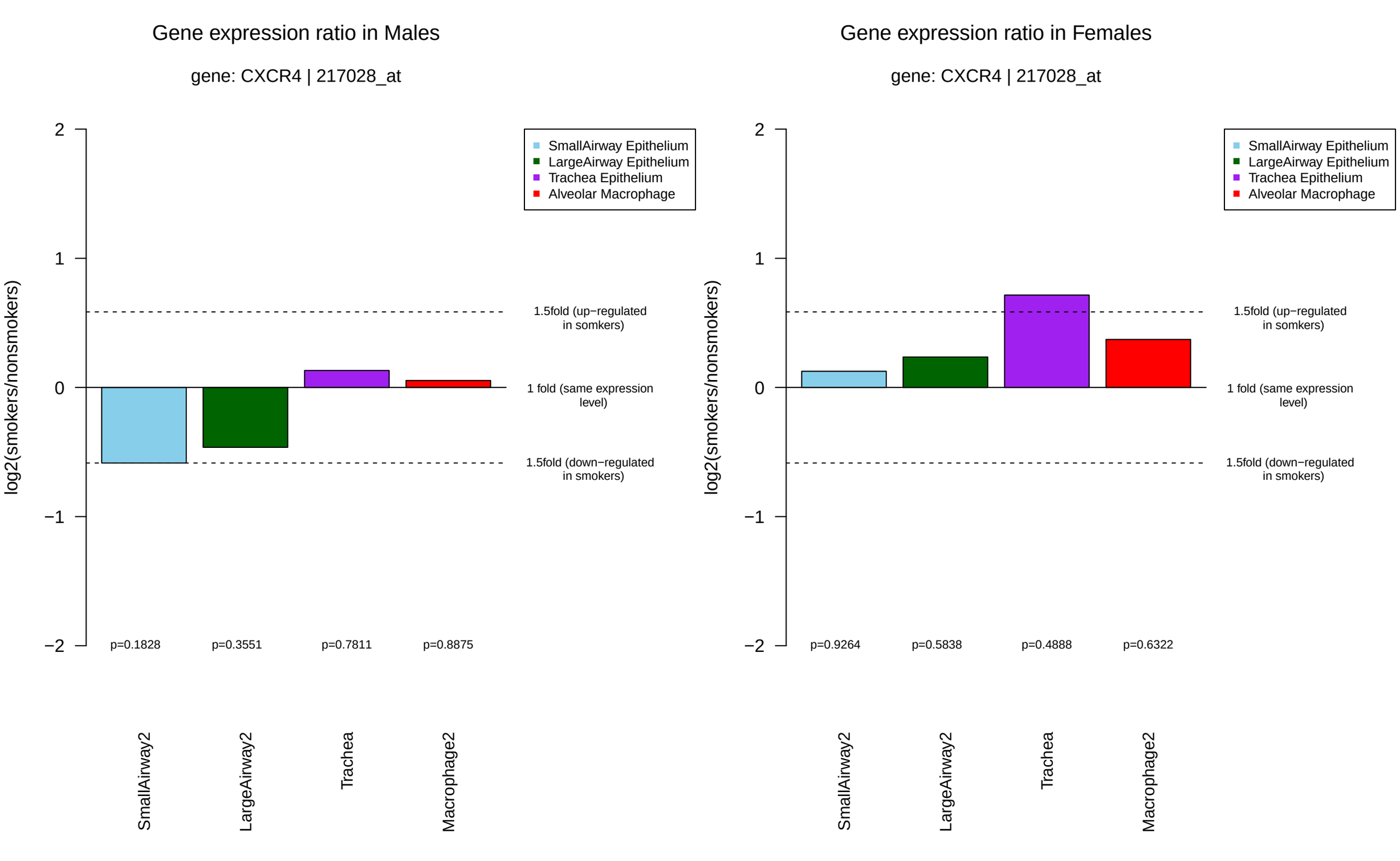
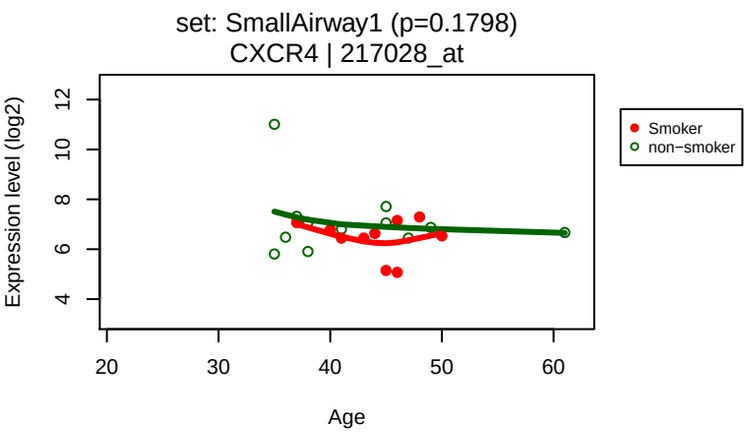
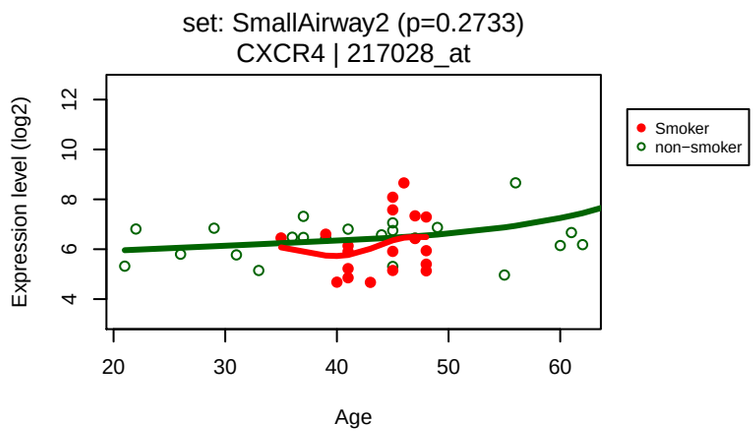
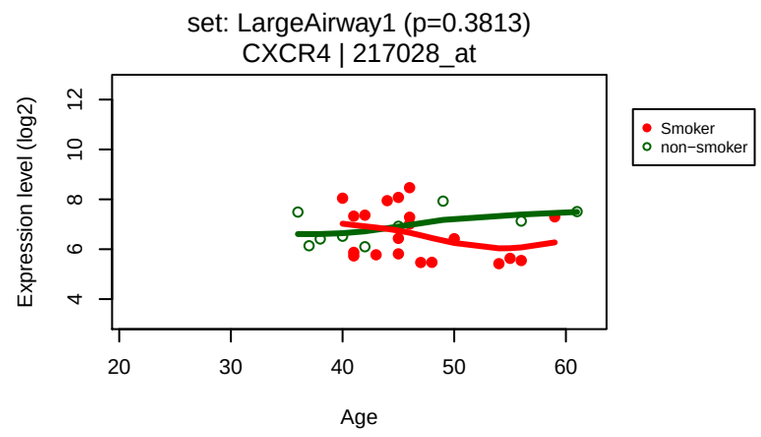

Gene expression of smokers and nonsmokers
gene: CXCR4 | 217028_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1798
p=0.2733
p=0.0342
p=0.3813
p=0.4288
p=0.4476
p=0.2951
p=0.5906
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

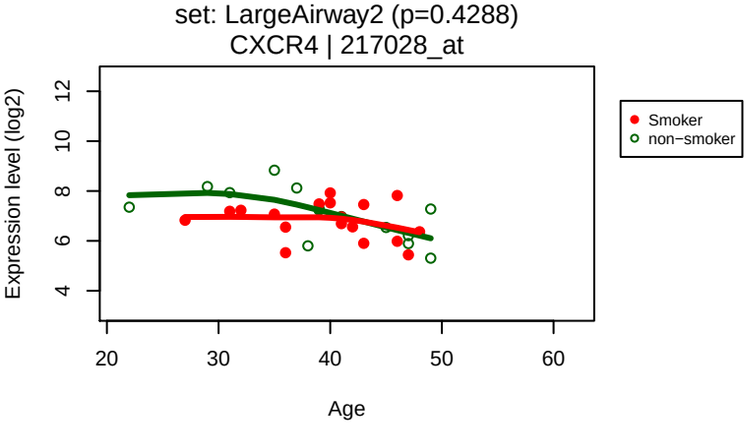

Gene expression ratio (smokers/nonsmokers)
gene: CXCR4 | 217028_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1798
p=0.2733
p=0.0342
p=0.3813
p=0.4288
p=0.4476
p=0.2951
p=0.5906
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

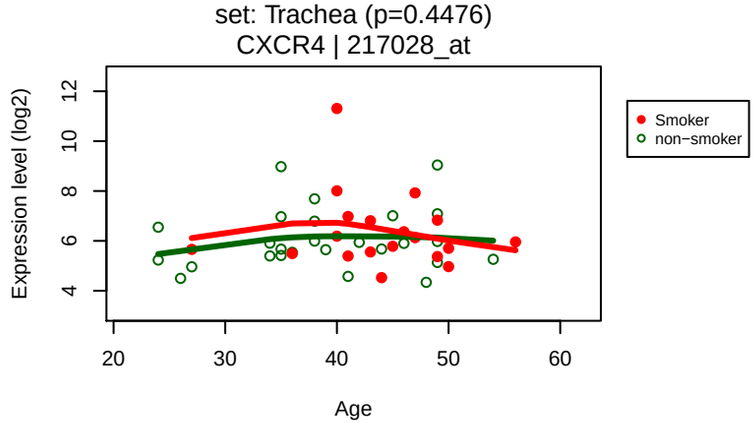

Gene expression ratio in Males
gene: CXCR4 | 217028_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1828
p=0.3551
p=0.7811
p=0.8875
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CXCR4 | 217028_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9264
p=0.5838
p=0.4888
p=0.6322
SmallAirway2
LargeAirway2
Trachea
Macrophage2

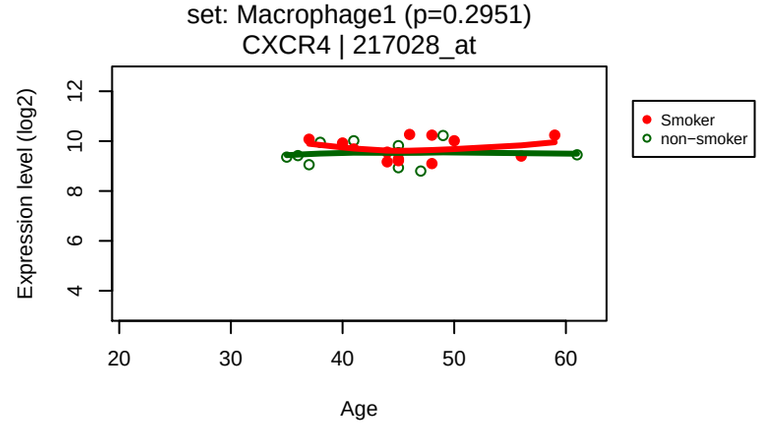

set: SmallAirway1 (p=0.1798)
CXCR4 | 217028_at
Expression level (log2)
Age
Smoker
non−smoker

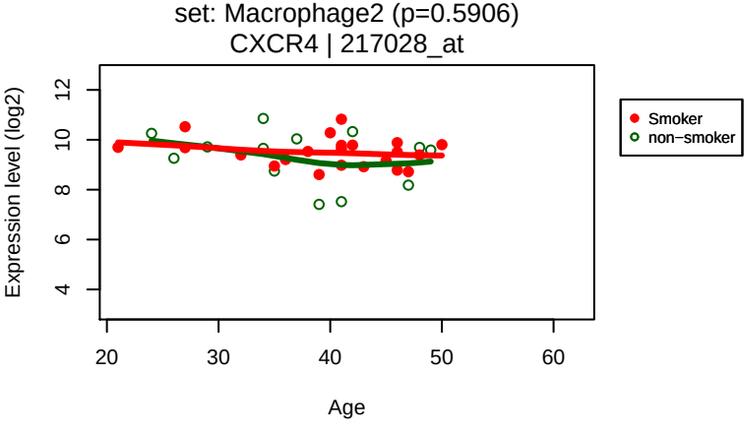

set: SmallAirway2 (p=0.2733)
CXCR4 | 217028_at
Expression level (log2)
Age
Smoker
non−smoker

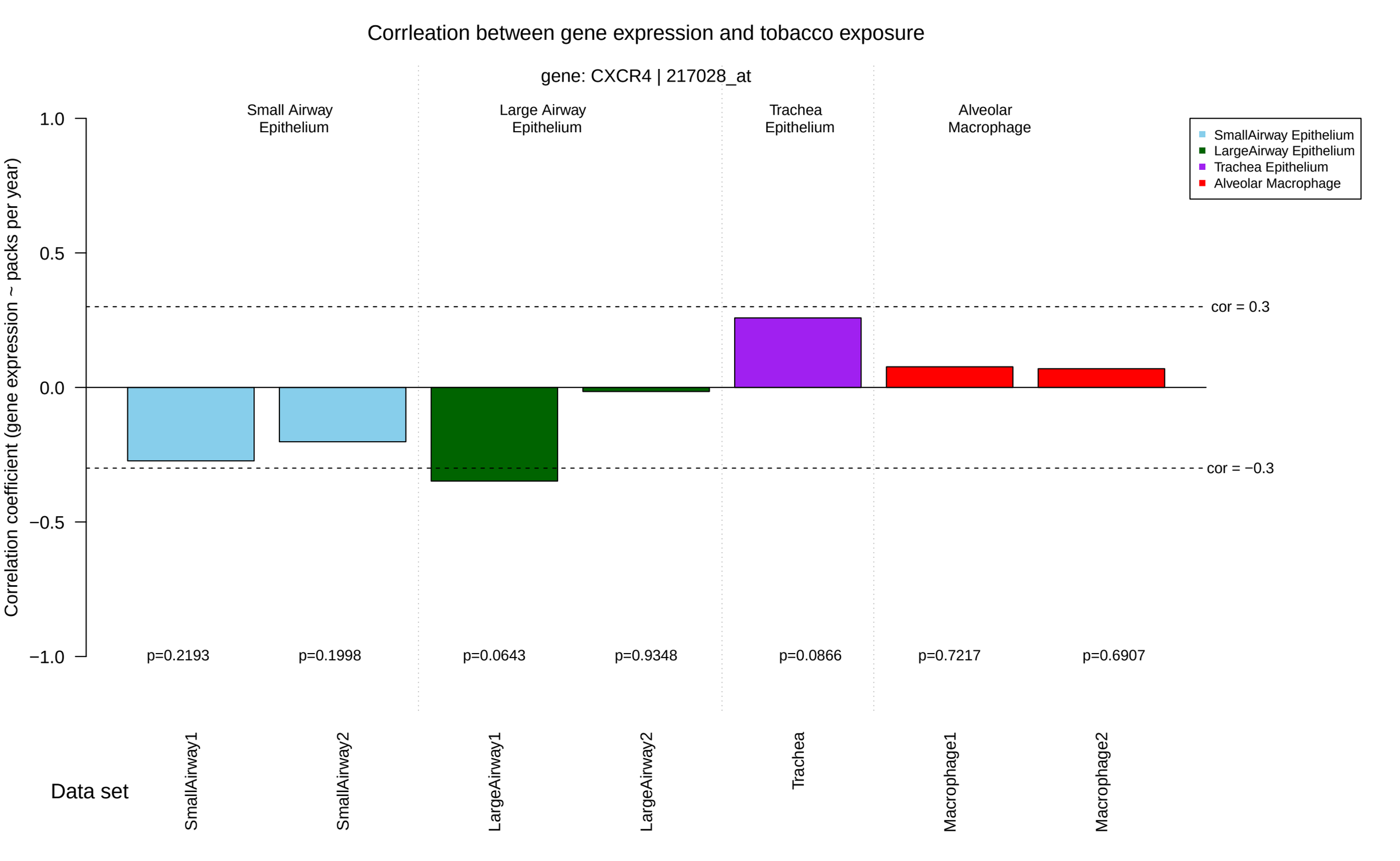

set: LargeAirway1 (p=0.3813)
CXCR4 | 217028_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4288)
CXCR4 | 217028_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4476)
CXCR4 | 217028_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2951)
CXCR4 | 217028_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5906)
CXCR4 | 217028_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CXCR4 | 217028_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2193
p=0.1998
p=0.0643
p=0.9348
p=0.0866
p=0.7217
p=0.6907
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2